

Cyber Security Attack Analysis

A Data Analytics Project using Python, SQL, and Tableau

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Tools & Technologies: Python, Pandas, NumPy, Matplotlib, SQLite, Tableau, Git, GitHub

Dataset: Cyber Security Attack Dataset (Kaggle)

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1. Executive Summary

Cybersecurity has become a critical concern for organizations due to the increasing frequency and sophistication of cyber attacks. This project analyzes a cybersecurity attack dataset containing **40,000 network events** to identify attack patterns, protocol usage, severity levels, network segment activity, and anomaly trends.

The project demonstrates the complete data analytics workflow, including data cleaning, exploratory data analysis (EDA), SQL-based querying, and interactive dashboard development using Tableau. The objective is to transform raw cybersecurity data into meaningful business insights that support effective threat monitoring and decision-making.

2. Problem Statement

Organizations generate massive volumes of network security logs every day. Identifying attack patterns and understanding network behavior manually is difficult and time-consuming.

This project aims to analyze cybersecurity attack data to answer key business questions, including:

- What types of cyber attacks occur most frequently?
- How are attacks distributed across network segments?
- Which protocols are commonly associated with attacks?
- How severe are the recorded attacks?
- What actions are taken against different attack types?

3. Project Objectives

The primary objectives of this project are:

- Load and understand the cybersecurity dataset.
- Clean and preprocess the data.
- Perform Exploratory Data Analysis (EDA).
- Analyze data using SQL queries.
- Develop interactive Tableau dashboards.
- Generate business insights and recommendations.

4. Dataset Description

Attribute	Details
Dataset Source	Kaggle
Number of Records	40,000
Number of Features	25
Data Type	Structured
Domain	Cyber Security

Dataset Overview

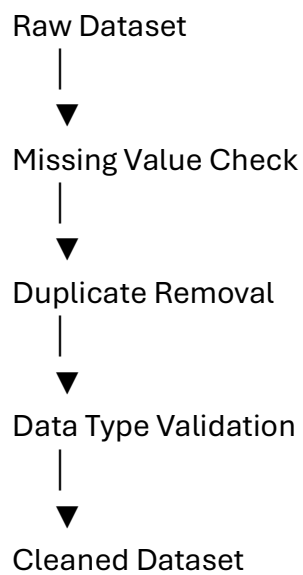
- Timestamp
- Source IP Address
- Destination IP Address
- Source Port
- Destination Port
- Protocol
- Packet Length
- Packet Type
- Traffic Type
- Attack Type
- Severity Level
- Action Taken
- Network Segment
- Geo-location Data
- Anomaly Scores

5. Data Cleaning & Preprocessing

The following preprocessing steps were performed:

- Removed duplicate records.
- Checked and handled missing values.
- Verified and corrected data types.
- Converted Timestamp into DateTime format.
- Removed unnecessary inconsistencies.
- Saved the cleaned dataset for further analysis.

Data Cleaning Workflow



6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Python libraries such as Pandas and Matplotlib.

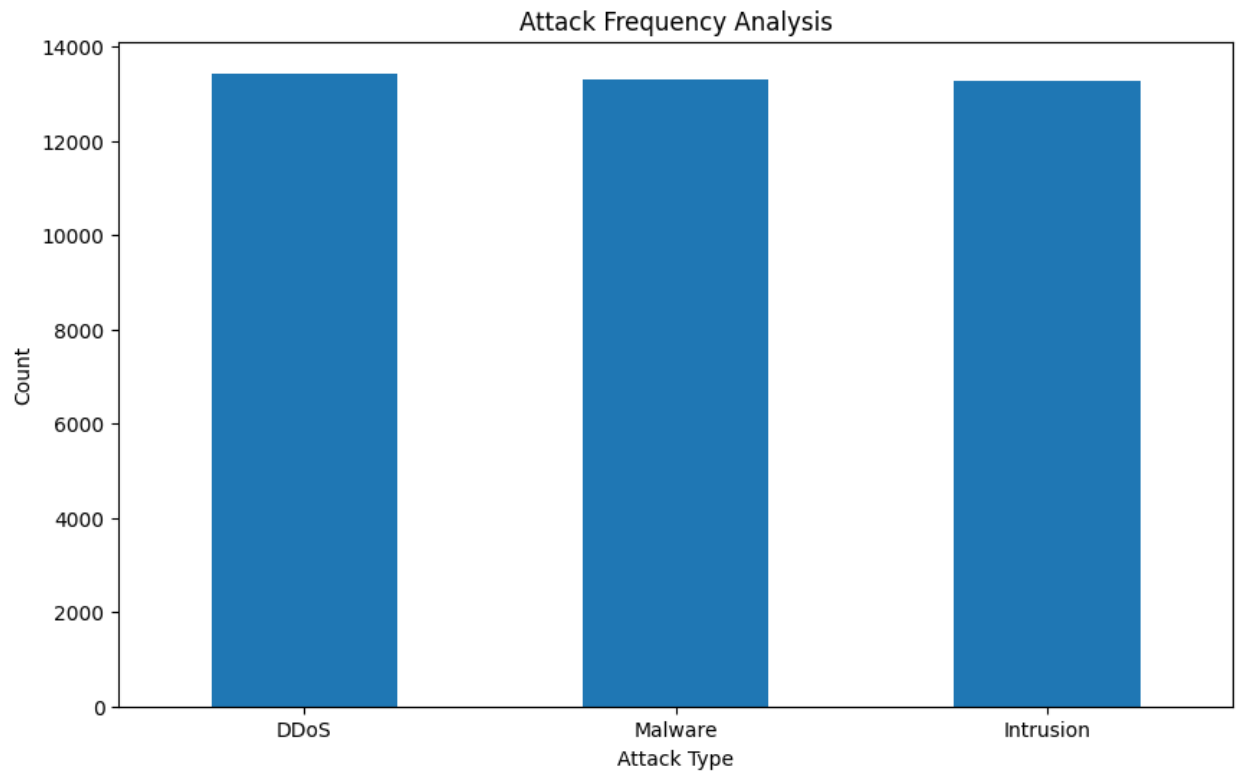
The following analyses were conducted:

- Attack Type Distribution
- Severity Level Distribution
- Protocol Distribution
- Action Taken Analysis
- Monthly Attack Trend

- Network Segment Analysis
- Geo-location Analysis
- Correlation Analysis

EDA Visualizations

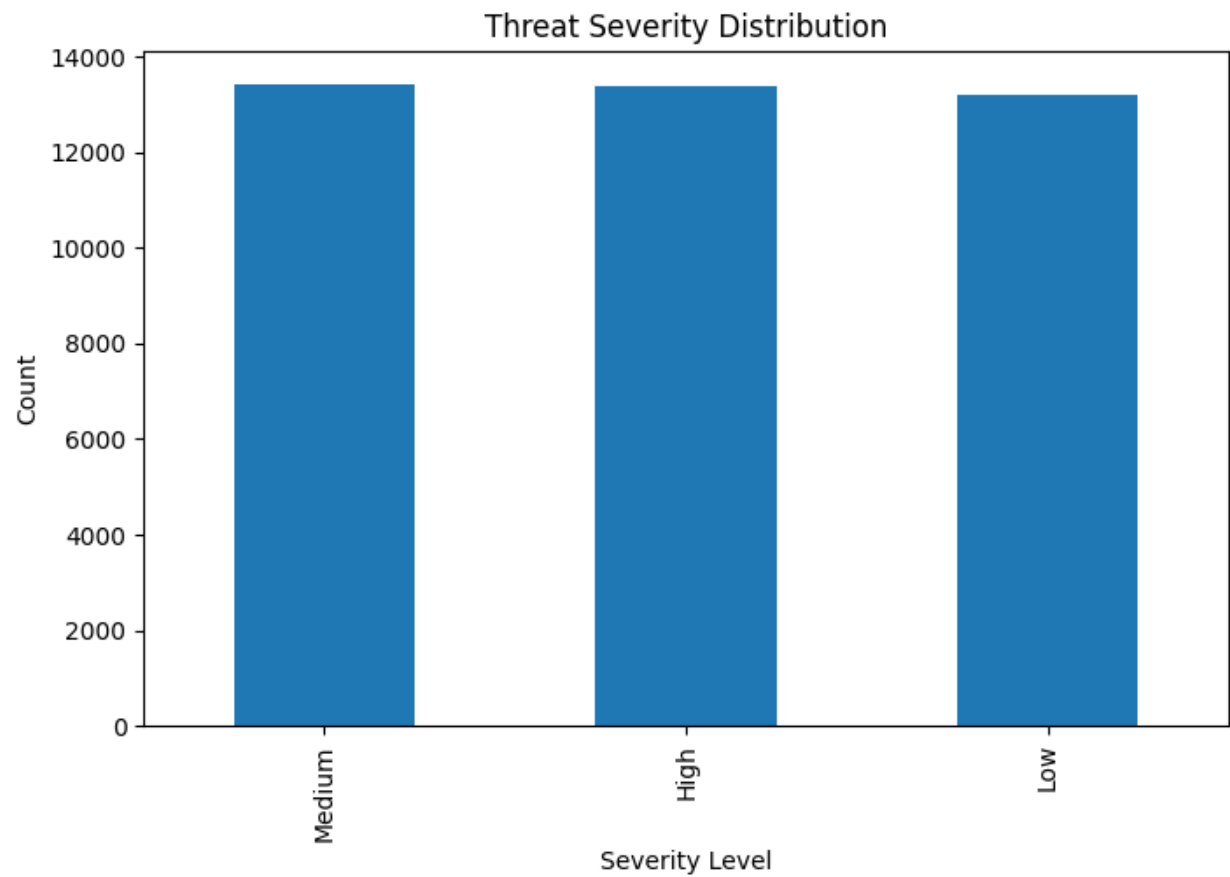
Attack Type Distribution



Business Insight

Attack types are almost evenly distributed, indicating multiple threat vectors rather than a single dominant attack type.

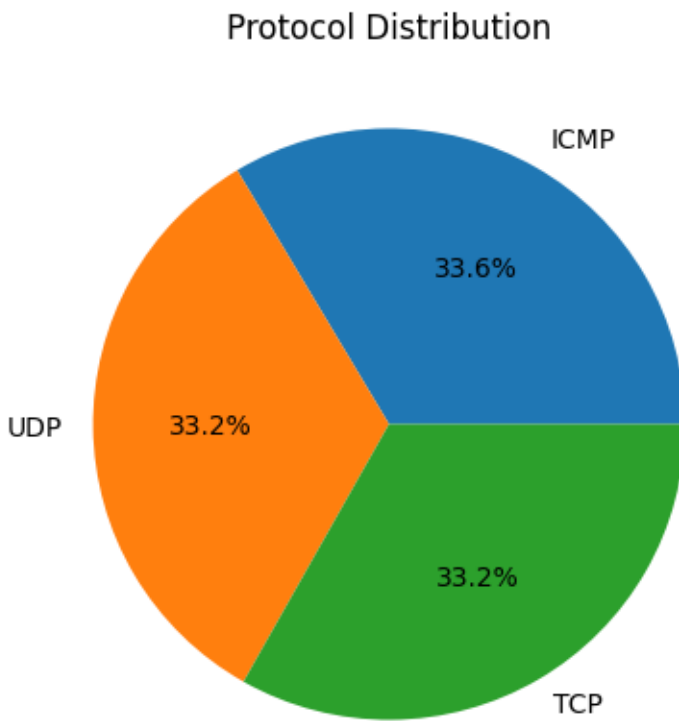
Severity Level Distribution



Business Insight

High, Medium, and Low severity attacks occur in comparable proportions, suggesting organizations should monitor all severity levels.

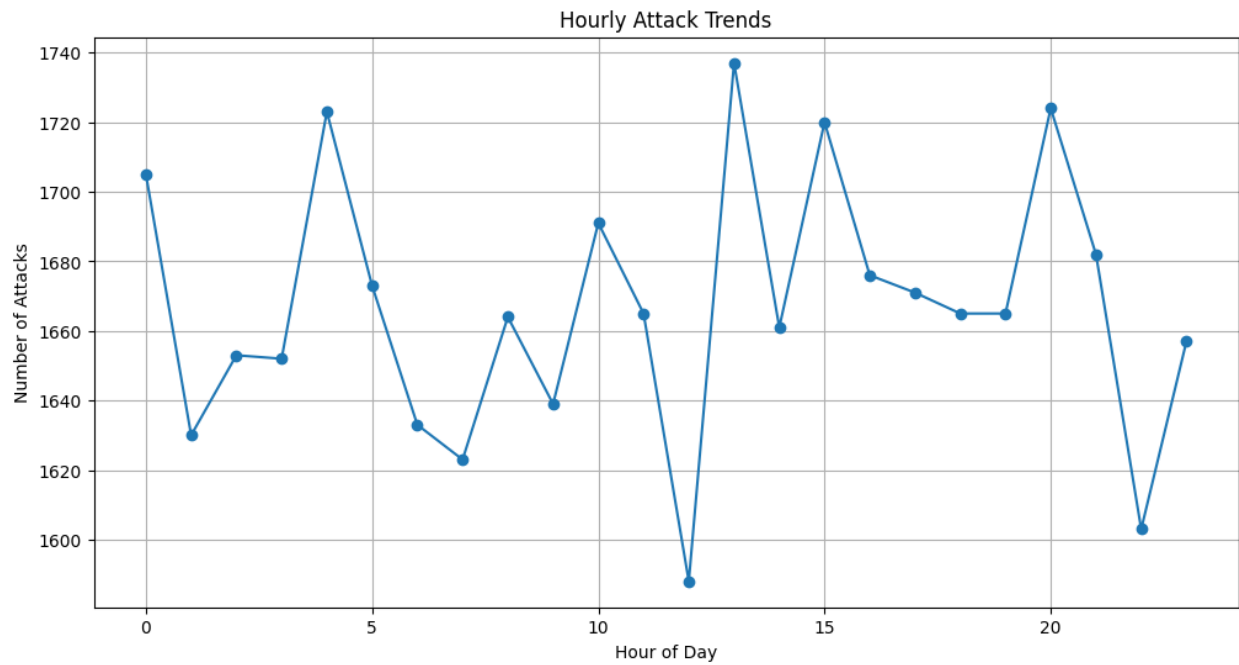
Protocol Distribution



Business Insight

TCP, UDP, and ICMP protocols exhibit balanced attack frequencies, indicating that attacks are not concentrated on a single protocol.

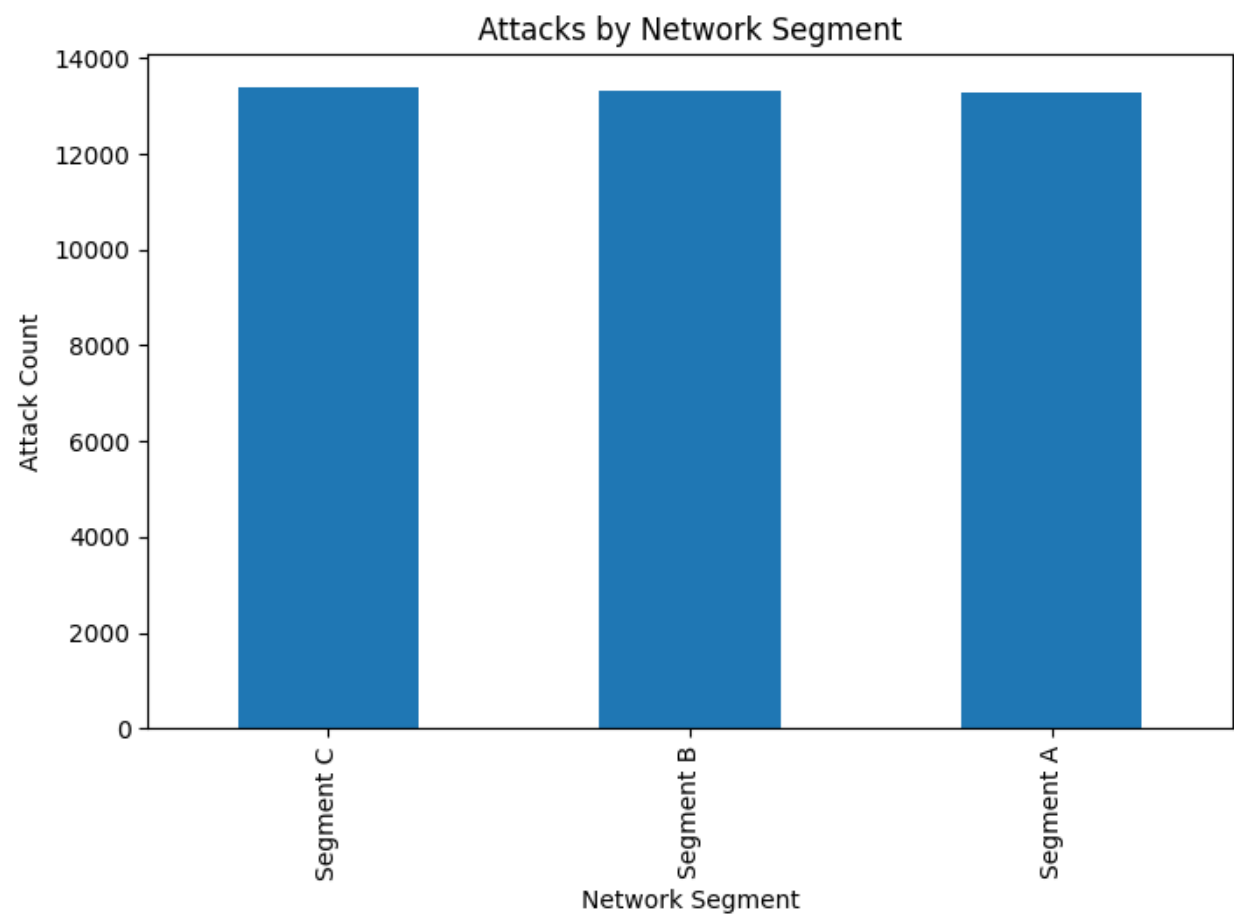
Hourly Attack Trend



Business Insight

Certain periods experience higher attack activity. Security monitoring and alert systems should be strengthened during peak attack hours. SOC resource allocation can be optimized using time-based attack analysis.

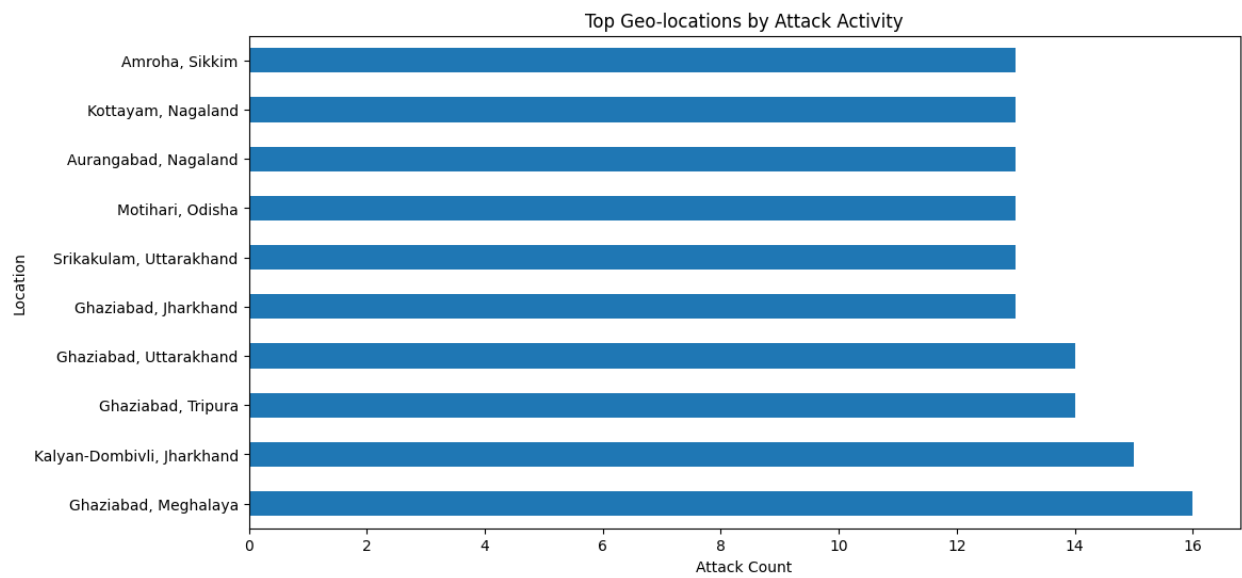
Network Segment Analysis



Business Insight

Attack volumes are evenly distributed across Segments A, B, and C, suggesting that all network segments require consistent security monitoring.

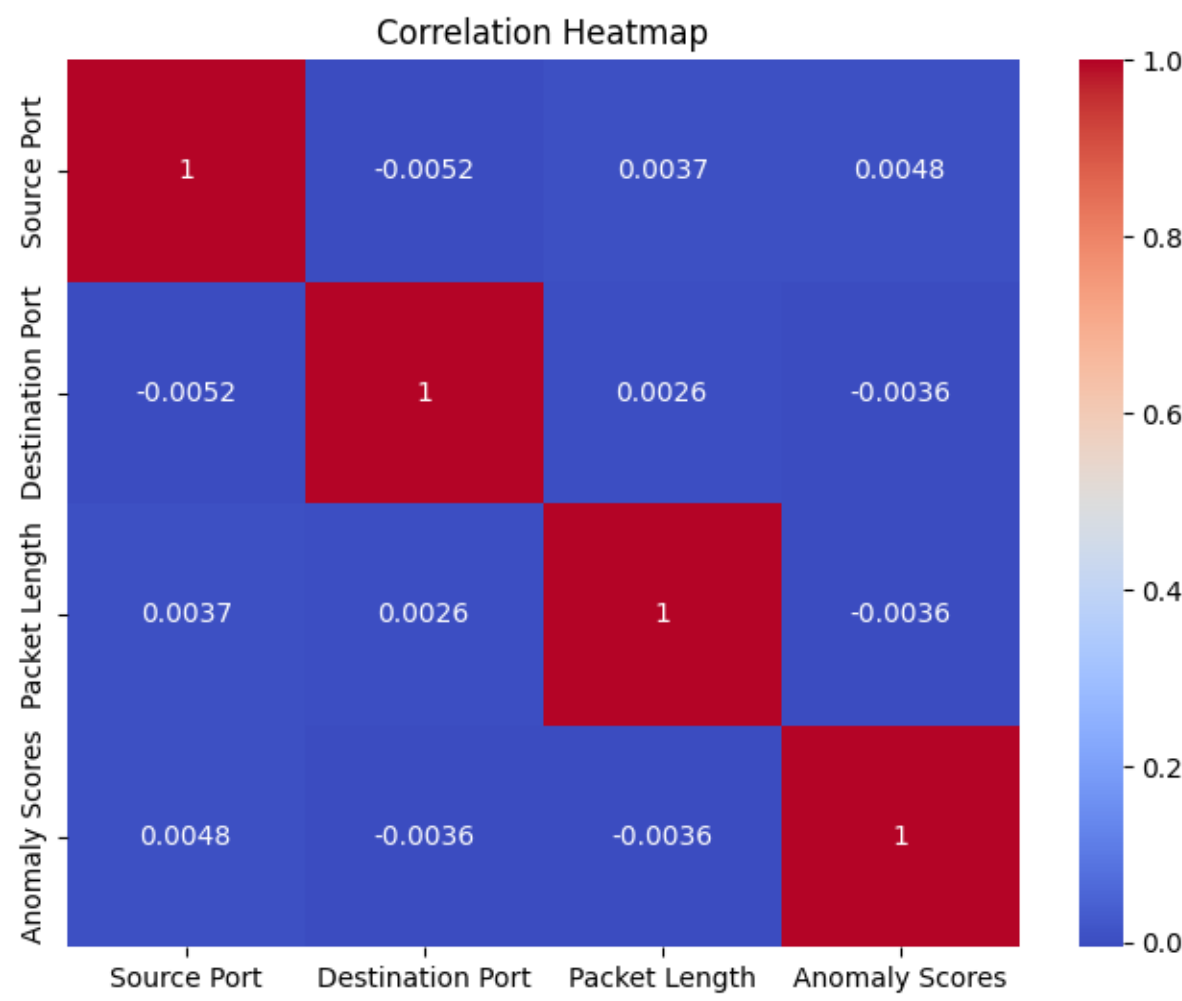
Geo-location Analysis



Business Insight

Attack events are geographically dispersed, with no single location dominating attack activity.

Correlation Matrix



Business Insight

Numerical features exhibit very weak correlations, indicating that variables such as packet length, ports, and anomaly scores operate largely independently.

7. SQL Analysis

SQLite was used to perform SQL-based analysis on the cleaned dataset.

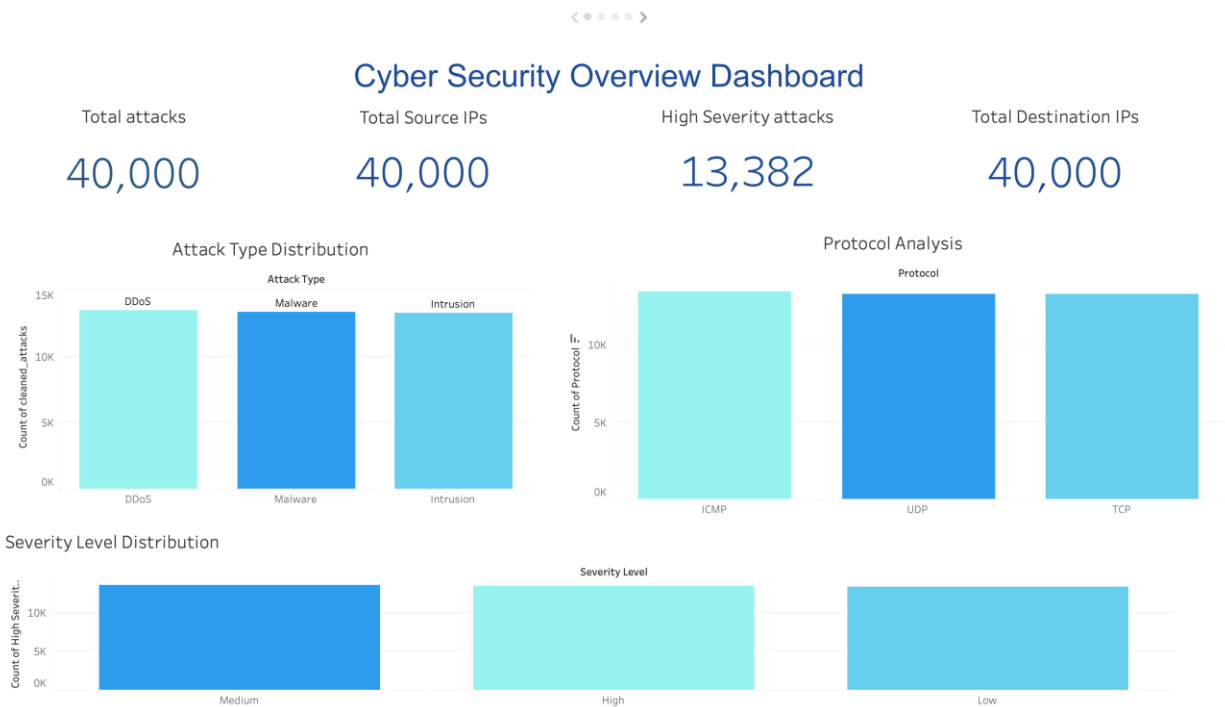
The following analyses were performed:

- Attack count by attack type
- Severity level distribution
- Protocol analysis
- Network segment analysis
- Action taken analysis
- Average anomaly score
- Monthly attack trend

8. Tableau Dashboard

Interactive dashboards were developed using Tableau to provide meaningful business insights.

Dashboard 1 – Cyber Security Overview



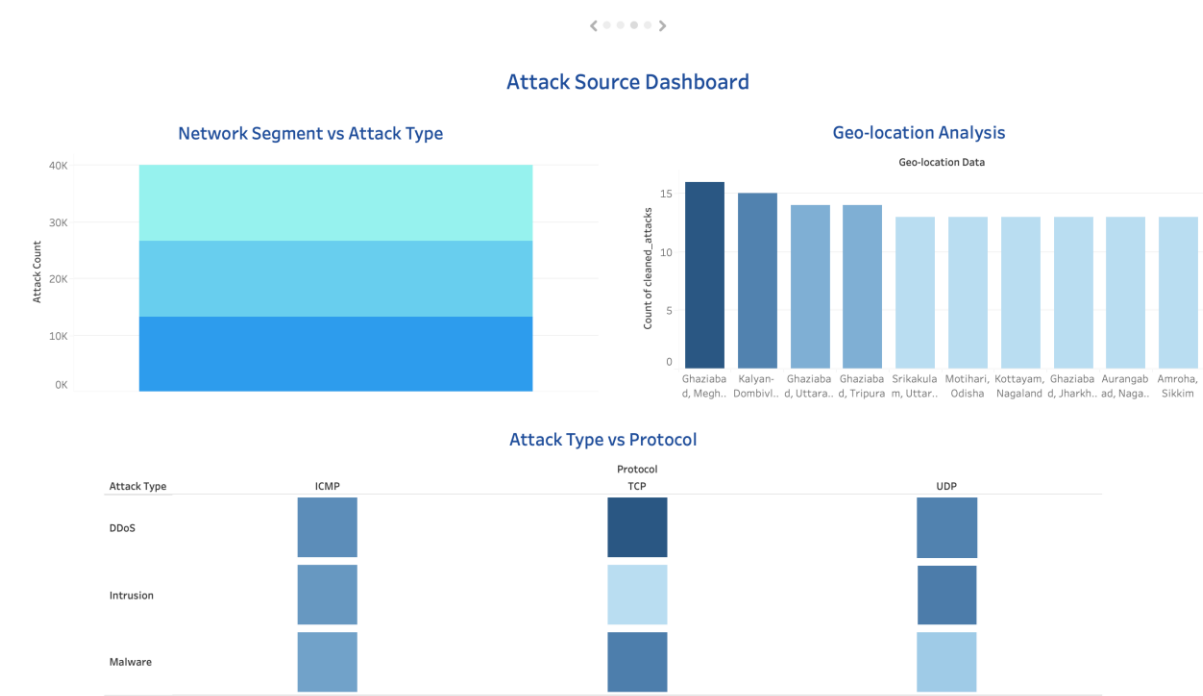
Purpose

Provides an executive summary of cybersecurity activity.

Key Visualizations

- KPI Cards
- Attack Type Distribution
- Severity Distribution
- Protocol Distribution
- Monthly Trend

Dashboard 2 – Attack Source Dashboard



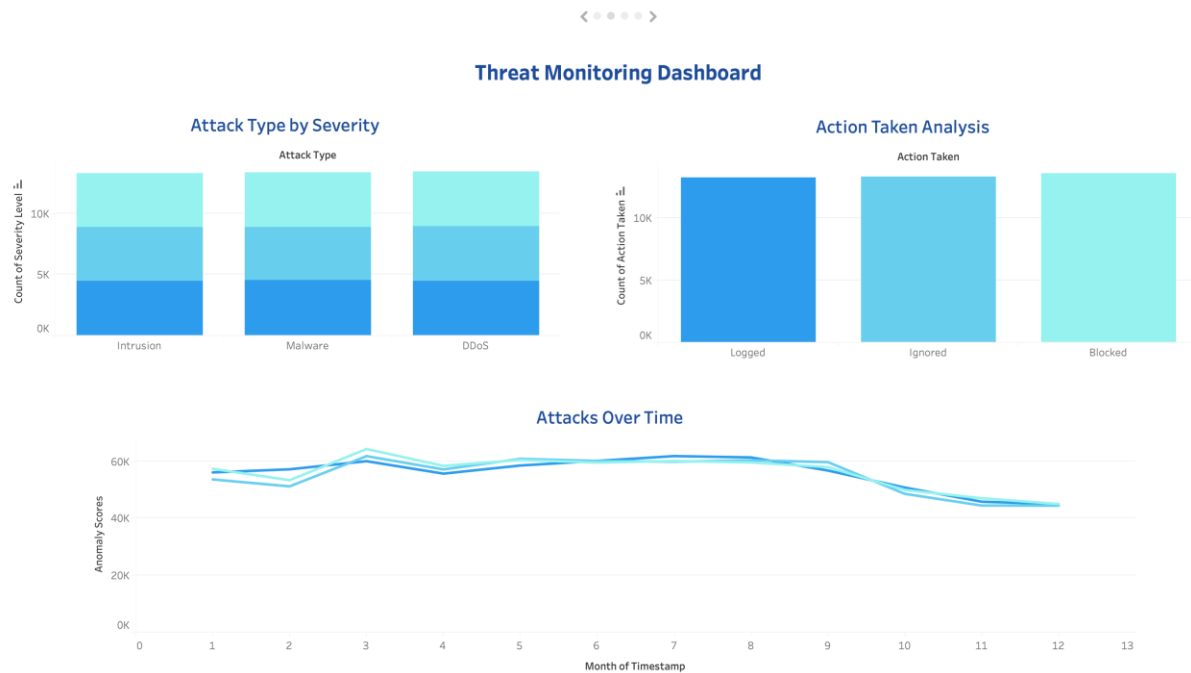
Purpose

Analyzes the distribution of attacks across network segments, geographic locations, and communication protocols.

Key Visualizations

- Geo-location Analysis
- Network Segment Analysis
- Attack Type vs Protocol Heatmap

Dashboard 3 – Threat Monitoring Dashboard



Purpose

Monitors attack severity, trends, and response actions over time.

Key Visualizations

- Monthly Attack Trend
- Attack Type vs Severity
- Action Taken Analysis
- Protocol vs Severity

Dashboard 4 – Risk Severity Dashboard



Purpose

Evaluates cybersecurity risk using anomaly scores and severity levels.

Key Visualizations

- Average Anomaly Score by Attack Type
- Traffic Status (Normal vs Suspicious)
- Severity by Network Segment
- Average Anomaly Score by Protocol

9. Key Findings

- The dataset contains **40,000 cybersecurity attack records**.
- Attack types are distributed almost evenly across categories.
- Severity levels show balanced distributions of High, Medium, and Low risk.
- Network segments experience similar attack volumes.
- TCP, UDP, and ICMP protocols are all common attack vectors.
- Numerical features exhibit minimal correlation, suggesting independent attack characteristics.
- Attack activity remains relatively stable over time.
- Interactive dashboards improve the monitoring and interpretation of cybersecurity events.

10. Business Recommendations

- Continuously monitor all network segments, as attack activity is evenly distributed.
- Prioritize investigation of high-severity incidents.
- Implement automated monitoring for high anomaly scores.
- Regularly review protocol-specific attack patterns.
- Enhance dashboard monitoring for real-time threat detection.
- Integrate machine learning techniques for predictive anomaly detection.

11. Conclusion

This project successfully demonstrated an end-to-end cybersecurity data analysis workflow using Python, SQLite, and Tableau. The analysis transformed raw network security data into meaningful insights through data cleaning, exploratory analysis, SQL querying, and interactive dashboards.

The developed dashboards provide a clear understanding of attack patterns, protocol usage, severity levels, and network behavior, supporting informed decision-making in cybersecurity operations.

12. Future Scope

Future enhancements may include:

- Real-time cybersecurity dashboard integration.
- Machine learning models for attack prediction.
- Automated anomaly detection systems.
- Cloud deployment using AWS or Azure.
- Integration with Security Information and Event Management (SIEM) platforms.

13. References

- Kaggle – Cyber Security Attack Dataset
- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- SQLite Documentation
- Tableau Documentation